

Take-Home Assignment

Senior Python Engineer (contract) • nymble health

Thanks for making it this far. This is the last major step before we decide

To test that, we're going to hand you a real, working project and let you decide what to do to it.

The project

A working RAG system that answers questions grounded in healthcare product monographs — currently Lipitor and Metformin.

Repo <https://github.com/timpowellgit/healthcare-rag>

Click the green **"Use this template"** button on the repo, pick **Private** as the visibility, and work in the private repo it creates for you. Follow the README to get it running. Try that before you plan, the point is to have a real feel for the thing before you decide what to do to it.

Your task

Below are ten directions this project could be taken.

Direction 4 is required. We build in healthcare, and how you reason about what this app must refuse to do tells us more than anything else on the list.

Beyond that, pick two or three. Tell us why you picked those — and what you deliberately left alone, and why. Picking a direction we did not list is also fine.

- 1 Bring it up to the standards you'd expect of any code you'd inherit. Turn this from a working demo into something a small team could actually run.
- 2 Give this a face. Make it something a person would want to use, not something that still feels like a prototype.
- 3 Make it possible to know whether a change made things better or worse. Give it a safety net for regressions.
- 4 **REQUIRED** Take the healthcare context seriously. There are things this app should not do, data it should not touch, cases it should not answer. Figure out what those are and enforce them.
- 5 Question the approach itself. The framework, the orchestration, the pipeline shape, none of it is sacred. If there's a simpler or better way to do this job, build that.
- 6 Assume an AI teammate will be the next contributor, and prepare accordingly.
- 7 Ship it. Do the work between the code being written and a real person being able to use it right now.
- 8 Make it cheaper to run. Treat the API bill as a design constraint.
- 9 Make it survive when things go wrong. Anything can fail so make the app graceful about that.
- 10 Make it more abstract. Right now it does one specific job with one specific set of pieces. Turn it into something a lot more general.

What we require from your AI-coding process

We use AI coding tools heavily. We want to see how you do.

Push your tools as far as you can, and **bake what worked into the repo**. Whatever helped your AI collaborator work well on this project, i.e. a skill, a hook, an agents file, a note in memory, a project convention, an eval loop your tool could call. Or something much more sophisticated.

THE BAR WE'RE MEASURING AGAINST

If a stranger clones your fork tomorrow and turns on their own AI coding tool, they should be faster because you were here first.

Tell us about this explicitly in your PR description: which tools you used, how you drove them, what patterns worked, and what artefacts you left behind for the next AI teammate on this repo.

What to submit

Three things:

1 Your changes, merged into your `main`.

Use as many pull requests as makes sense as you go.

2 A technical write-up in a Google Doc

covering four things:

- **Your read on the project.** What you thought of it the first time you read it, what was working, what wasn't, what you were tempted to touch first.
- **The work itself.** For each direction you touched, a short section: what you built, why you built it that way, and what you'd change on a second pass.
- **Judgment.** The biggest trade-off you made. What surprised you along the way. If you had another week, what you'd do next.
- **Your AI-coding process.** Which tools you used and how you drove them. Patterns that worked, and patterns that didn't. Artefacts you left behind for the next AI teammate on this repo.

3 A five-minute video

walking us through what you changed and why. No need for polish — a screen recording with your voice over it is fine. Drop the link into the Google Doc so we can find everything from one place.

Keep your work private. Click the green **"Use this template"** button on the repo, pick **Private** as the visibility, and work in the private repo it creates for you. We may have several candidates working on this at the same time and we want each submission to stand on its own.

Once you're done

Grant [@timpowellgit](#) collaborator access on your private repo and share the Google Doc with insights@nymble.health (or wherever you want them to send it). Don't do either of these before you're done, as we'd rather see the finished submission cold.

Practical notes

- **Time.** About three days of focused effort. Spread it over a week if you need to. If it takes longer, tell us.
- **Questions.** Email us any time.

Good luck. Have fun with it.